MEASURING THE MOAT: EXCESS BOUNDS NEAR SEYMOUR'S SECOND NEIGHBOURHOOD CONJECTURE

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ABSTRACT. Seymour's second neighbourhood conjecture (SSNC) asserts that every oriented graph has a vertex whose second out-neighbourhood is at least as large as its first. For an oriented graph G define $\mathrm{exc}(G) = \sum_v \max(0,\, d_2(v) - d_1(v) + 1)$; then $\mathrm{exc}(G) = 0$ iff G is a counterexample. Let $E_{\geq d}(n)$ be the minimum excess over n-vertex oriented graphs of minimum out-degree at least d, and $E^{\mathrm{sc}}_{\geq d}(n)$ the same minimum over the strongly connected ones. We prove $E_{\geq 8}(n) \leq E^{\mathrm{sc}}_{\geq 8}(n) \leq 3 + [3 \nmid n]$ for every $n \geq 24$, by exhibiting one explicit family; this subsumes every upper bound of the previous version. At the opposite extreme we determine E exactly on a degenerate family: $E_{\geq k}(2k+1) = 2k+1$ for all $k \geq 1$, by an elementary argument that replaces a solver computation. Inside one explicitly defined class of capped-tournament blow-ups we prove a partial lower bound, $\mathrm{exc} \geq 3$ with equality forcing $3 \mid n$; we emphasise that this says nothing outside that class. No lower bound beyond the trivial $\mathrm{exc} \geq 0$ is known at any $n \geq 18$. We also record, as a methodological result, that search attainment overstated the best available upper bound in every discrepancy, and that in each case an explicit construction, not a further search, settled it.

1. INTRODUCTION

Seymour's second neighbourhood conjecture [DL95] states:

Conjecture 1.1 (SSNC)**.** *Every oriented graph (digraph without 2-cycles) contains a vertex v with $|N_2^+(v)| \geq |N_1^+(v)|$.*

The conjecture is proved for tournaments [Fis96, HT00] and in many special classes; every oriented graph has a vertex with $|N_2^+(v)| \geq \gamma\,|N_1^+(v)|$ for $\gamma = 0.715538\ldots$ [HP24], improving [CSY03]. A counterexample must have minimum out-degree $\delta^+ \geq 8$: out-degree ≤ 6 was eliminated by [KL01] and $\delta^+ = 7$ by [SSS26] via a signature-compression argument checked by CP-SAT.

Recent work studies the *equality boundary* of the conjecture. [Hal26] calls a strongly connected oriented graph **Pisa** if $\max_v(|N_2^+(v)| - |N_1^+(v)|) = 0$; [GKZ26] study **Seymour-tight** orientations, in which *every* vertex meets the bound, and develop a product calculus for them. In the tight case the

Version 2.1 — 2026-07-26. Supersedes version 1.1 (2026-07-23), which remains archived. All proved inequalities of that version remain valid; this version corrects notation, interpretation and provenance.

set of boundary vertices is everything; in a generic Pisa graph it can be much smaller. This note measures how small.

Definitions. For an oriented graph G write $d_1(v) = |N_1^+(v)|$ and $d_2(v) = |N_2^+(v)|$, where $N_2^+(v) = \{x : \mathrm{dist}^+(v,x) = 2\}$; so $N_1^+(v)$, $N_2^+(v)$ and $\{v\}$ are pairwise disjoint. Call $m(v) = d_2(v) - d_1(v)$ the *margin* of v, call v **tight** if $m(v) = 0$ and **satisfied** if $m(v) \geq 0$. Define

$$\mathrm{exc}(G) \;=\; \sum_{v \in V(G)} \max\bigl(0,\; m(v)+1\bigr).$$

Then $\mathrm{exc}(G) = 0$ iff G is a counterexample to SSNC, and for a Pisa graph the excess equals the number of tight vertices. Adding a universal sink produces exc $= 1$ trivially [GKZ26, §1], so the interesting quantity carries the degree bound that counterexamples must satisfy. *Two* minima are used here and they must not be conflated:

$$E_{\geq d}(n) = \min\{\mathrm{exc}(G) : |V(G)| = n,\; G \text{ oriented},\; \delta^+(G) \geq d\},$$
$$E^{\mathrm{sc}}_{\geq d}(n) = \text{the same minimum over the strongly connected such } G.$$

Since the second minimises over a subclass, $E_{\geq d}(n) \leq E^{\mathrm{sc}}_{\geq d}(n)$. Version 1.1 of this note defined only E^{sc}, writing it E, while the accompanying repository defined only E; the two were used interchangeably. They are separated here, and every statement below says which one it bounds. The reasons are uniform: an upper bound witnessed by a strongly connected graph bounds both, and an infeasibility argument run without a connectivity constraint rules out the larger class and hence the subclass as well. The restriction to strong connectivity is motivated by the minimal-counterexample reduction — a closed out-section of a counterexample is a smaller counterexample, cf. [EDGGK25].

$E_{\geq 8}(n) = 0$ for some n iff SSNC is false. If SSNC holds then $E_{\geq 8}(n) \geq 1$ for all n, and any stronger uniform lower bound is a strengthening of the conjecture.

Terminology: floor, bound, attainment. Version 1.1 and the repository README used the word “floor” for values reached by search. They are not floors, and the improvements reported here are what made that visible. Three words are kept apart throughout: *floor* means $E_{\geq d}(n)$ itself, the exact minimum — which we do not know at a single $n \geq 18$; *bound* means a proved inequality; and *attainment* means the best value some search actually reached, which is evidence for an upper bound and never for a lower one. Section 5.2 is the record of what happens when the three are mixed.

Contributions.

(1) **A flat upper bound** (§2): one explicit family gives $E^{\mathrm{sc}}_{\geq 8}(n) \leq 3 + [3 \nmid n]$ for every $n \geq 24$ (Theorem 2.6). It subsumes the pure-ring, cycle-power and keystone-transplant bounds of version 1.1, which are retained as provenance in §2.3.

(2) **An exact value on a degenerate family** (§3): $E_{\geq k}(2k+1) = E^{\mathrm{sc}}_{\geq k}(2k+1) = 2k+1$ for every $k \geq 1$ (Proposition 3.1), by six lines of argument. The $k = 8$ case supersedes the computer-assisted $E(17) \geq 3$ of version 1.1.
(3) **A class-limited partial lower bound** (§4, Appendix B): inside one explicitly defined family of capped-tournament blow-ups, exc $\geq$ 3, with equality forcing $3 \mid n$. This is *not* a lower bound on $E_{\geq 8}$.
(4) **Measurements and a measurement-theoretic caveat** (§5): search attainment failed in two distinct ways, and construction alone adjudicated each time.
(5) **Open problems** (§7), foremost: is $3 + [3 \nmid n]$ exact? No positive lower bound is known at any $n \geq 18$.

Verification protocol. Every witness is stored in the accompanying repository as an adjacency list with a SHA-1 fingerprint, and every numerical claim is checked by implementations written independently of one another: a numpy pipeline used by the search, a pure-Python verifier, and a standard-library-only verifier written from the definitions without access to the other two. A `make verify-*` target reproduces each claim, and every claim in the repository's table has one. The solver logs are *not* uniform in provenance; §3.4 states exactly which fields each records.

2. The flat upper bound

2.1. **The out-set inclusion lemma.** All constructions here are perturbed blow-ups of directed cycles. The mechanism that keeps perturbations from leaking is the following.

Lemma 2.1 (out-set inclusion). *Let G be an oriented graph and let $v \neq p$ be non-adjacent vertices with $N_1^+(p) \subseteq N_1^+(v)$. Let $G' = G + (v \to p)$. Then G' is oriented, and:*

(i) *if $p \in N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 2$;*
(ii) *if $p \notin N_1^+(G, v) \cup N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 1$;*
(iii) *for $x \neq v$, margins change only by the possible appearance of p in $N_2^+(x)$ through the new path $x \to v \to p$: precisely, $m_{G'}(x) = m_G(x) + 1$ exactly when $v \in N_1^+(x)$ and $p \notin N_1^+(x) \cup \{x\} \cup N_2^+(G, x)$; all other margins are unchanged. In particular, if every in-neighbour x of v satisfies $p \in N_1^+(x)$, then no margin other than $m(v)$ changes.*

Proof. G' is oriented since $p \to v$ is absent by non-adjacency. For v: $d_1(v)$ increases by exactly one. Candidate new second neighbours of v are the out-neighbours of the new first neighbour p, i.e. $N_1^+(p) \setminus (N_1^+(G', v) \cup \{v\}) = \emptyset$ by hypothesis; and no old second neighbour $u \neq p$ is lost, since arcs are only added and $u \notin N_1^+(G', v)$. The only possible change to $N_2^+(v)$ is the removal of p itself, which occurs precisely in case (i); this gives (i) and (ii). For $x \notin \{v, p\}$: $d_1(x)$ is unchanged, and the only new directed paths of length two are of the form $x \to v \to p$, which affect membership of p alone;

the stated condition characterises when p actually enters $N_2^+(x)$. For $x = p$: no new path leaves p, and no new path of length two into $N_2^+(p)$ is created, since such a path would require an arc into v to be new. ☐

In words: *if the target's out-set is already inside the shooter's first neighbourhood, the shot costs the shooter one unit of margin and is invisible to bystanders who see both.* The same arithmetic underlies the in-star trick in the generalized lexicographic products of [GKZ26].

2.2. **The family G_n.** Fix $n \geq 24$ and write $m = \lfloor n/3 \rfloor$, $r = n \bmod 3$. Partition V into three layers L_0, L_1, L_2 of sizes

$$s_a \;=\; m + [\, a < r \,] \qquad (a = 0, 1, 2),$$

so $s_0 + s_1 + s_2 = n$. Layers are consecutive blocks: with $\mathrm{st}_a = s_0 + \cdots + s_{a-1}$, the j-th vertex of L_a is $u_{a,j} = \mathrm{st}_a + j$. Indices of layers are mod 3. Put

$$\Delta_a \;=\; s_{a+2} - s_{a+1}, \qquad c_a \;=\; \max(0,\, \Delta_a + 1).$$

The arcs of G_n are exactly

(R) $u_{a,j} \to w$ for every $w \in L_{a+1}$ (complete, forward);
(T) $u_{a,j} \to u_{a,l}$ for every $l < \min(j, c_a)$ (an intra-layer transitive tournament, truncated at the cap c_a).

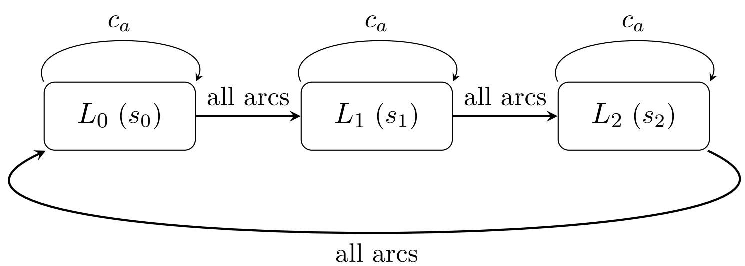

FIGURE 1. The family G_n. Layer sizes are $s_a = \lfloor n/3 \rfloor + [a < r]$, so they differ by at most one. Between layers, every arc runs forward; inside a layer, a transitive tournament truncated at the cap c_a. For $v \in L_a$, $N_1^+(v)$ is L_{a+1} together with at most c_a same-layer vertices, while $N_2^+(v) = L_{a+2}$ exactly (Lemma 2.3) — the intra-layer arcs raise d_1 without ever enlarging N_2^+.

Lemma 2.2 (well-formedness)**.** *G_n is an oriented graph, is strongly connected, and has $\delta^+(G_n) = \lfloor n/3 \rfloor$.*

Proof. No loops: (T) requires $l < \min(j, c_a) \leq j$. No digons: (R) arcs run $L_a \to L_{a+1}$, and a reverse arc would need $a + 2 \equiv a \pmod 3$; (T) arcs run strictly downwards in j inside one layer. An (R) arc and a (T) arc never join the same pair. Out-degrees: $\deg^+(u_{a,j}) = s_{a+1} + \min(j, c_a) \geq s_{a+1} \geq m$, with equality at $j = 0$ in a layer of minimum size, so $\delta^+ = m = \lfloor n/3 \rfloor$, which is ≥ 8 exactly when $n \geq 24$. Strong connectivity: from any $u \in L_a$ every vertex of L_{a+1} is reached in one (R) step, hence every vertex of L_{a+2} in two and every vertex of L_a in three. $\square$

Lemma 2.3 (second neighbourhood)**.** *$N_2^+(u_{a,j}) = L_{a+2}$, hence $d_2(u_{a,j}) = s_{a+2}$.*

Proof. By definition

$$N_2^+(v) \;=\; \Big(\bigcup_{x\in N_1^+(v)} N_1^+(x)\Big) \setminus \big(N_1^+(v) \cup \{v\}\big),$$

so it suffices to compute the union and then delete. Write $p = \min(j, c_a)$, so $N_1^+(u_{a,j}) = L_{a+1} \cup \{u_{a,0}, \ldots, u_{a,p-1}\}$. Every $x \in L_{a+1}$ contributes $N_1^+(x) = L_{a+2} \cup$(some vertices of L_{a+1}); L_{a+1} is non-empty and each such x covers all of L_{a+2}, so the union contains L_{a+2} and nothing outside $L_{a+2} \cup L_{a+1}$. Every $u_{a,l}$ with $l < p \leq c_a$ contributes $N_1^+(u_{a,l}) = L_{a+1} \cup \{u_{a,0}, \ldots, u_{a,l-1}\}$ (using $l < c_a$), which is contained in $N_1^+(u_{a,j})$ since $l-1 < j$. So the two-step reach is $L_{a+2} \cup L_{a+1} \cup$ (a subset of N_1^+). Deleting $N_1^+(u_{a,j})$, which contains L_{a+1}, and the vertex $u_{a,j} \in L_a$ leaves exactly L_{a+2}, since L_{a+2} is disjoint from $L_a \cup L_{a+1}$. $\square$

This is the mechanism of Lemma 2.1 in bulk: the intra-layer targets are chosen so that their out-sets already lie inside the observer's first neighbourhood, so each (T) arc lowers $d_1 - d_2$ by exactly one without ever enlarging N_2^+.

Lemma 2.4 (margins and per-layer excess)**.** *$m(u_{a,j}) = \Delta_a - \min(j, c_a)$. Consequently, writing $T(x) = (x+1)(x+2)/2$ for $x \geq 0$ and $T(x) = 0$ for $x < 0$,*

$$\sum_{j=0}^{s_a-1} \max\big(0,\, m(u_{a,j}) + 1\big) \;=\; T(\Delta_a).$$

Proof. The first claim is Lemma 2.3 together with $d_1(u_{a,j}) = s_{a+1} + \min(j, c_a)$. For the sum: when $j \geq c_a$ the margin is $\Delta_a - c_a \leq -1$ and contributes 0; when $j < c_a$ it is $\Delta_a - j$, contributing $\max(0, \Delta_a + 1 - j)$, which vanishes once $j > \Delta_a$. Here $\Delta_a \in \{-1, 0, 1\}$ by Lemma 2.5 below, and $s_a \geq 8 > \Delta_a$, so all indices $j = 0, \ldots, \Delta_a$ exist and lie below the cap; the sum is $\sum_{j=0}^{\Delta_a} (\Delta_a + 1 - j) = T(\Delta_a)$ when $\Delta_a \geq 0$, and 0 when $\Delta_a < 0$. $\square$

Lemma 2.5 (imbalance profile)**.** *From $s_a = m + [a < r]$,*

$$\begin{array}{lll} r = 0: & (s_0, s_1, s_2) = (m, m, m), & (\Delta_0, \Delta_1, \Delta_2) = (0,0,0); \\ r = 1: & (m{+}1, m, m), & (0, +1, -1); \\ r = 2: & (m{+}1, m{+}1, m), & (-1, +1, 0). \end{array}$$

Proof. Direct substitution into $\Delta_a = s_{a+2} - s_{a+1}$. Note $\Delta_0 + \Delta_1 + \Delta_2 = 0$ always. $\square$

Theorem 2.6 (flat upper bound)**.** *For every $n \geq 24$, the graph G_n is oriented, strongly connected, has $\delta^+ = \lfloor n/3 \rfloor \geq 8$, and*

$$\mathrm{exc}(G_n) \;=\; 3 + [\,3 \nmid n\,].$$

Consequently $E_{\geq 8}(n) \leq E^{\mathrm{sc}}_{\geq 8}(n) \leq 3 + [3 \nmid n]$ for every $n \geq 24$.

Proof. By Lemmas 2.4 and 2.5, $\mathrm{exc}(G_n) = \sum_a T(\Delta_a)$ with $T(-1) = 0$, $T(0) = 1$, $T(1) = 3$. For $r = 0$ this is $1 + 1 + 1 = 3$; for $r = 1$ it is $1 + 3 + 0 = 4$; for $r = 2$ it is $0 + 3 + 1 = 4$. Lemma 2.2 gives the remaining assertions. ☐

Remark 2.7 (the bound does not need $\delta = 8$). $\delta^+(G_n) = \lfloor n/3 \rfloor$, so the same family gives $E^{\mathrm{sc}}_{\geq D}(n) \leq 3 + [3 \nmid n]$ for every $D \leq \lfloor n/3 \rfloor$. The threshold 8 is not what the construction is responding to.

Remark 2.8 (the Pisa reading is limited to $3 \mid n$). When $3 \mid n$ all three Δ_a vanish and G_n has three tight vertices and no vertex of positive margin, so it is Pisa and its excess counts tight vertices. When $3 \nmid n$ some layer has $\Delta_a = +1$ and contributes a vertex of margin $+1$ (excess $4 = 2 + 1 + 1$); such a G_n is not Pisa. So the reading of $E_{\geq d}$ as "the minimum number of margin-0 vertices" is a statement about the subsequence $3 \mid n$, not about $E_{\geq d}$ in general.

2.3. **The constructions of version 1.1, and what became of them.**

Version 1.1 proved $E^{\mathrm{sc}}(mt) \leq m$ for $m \geq 3$, $t \geq 8$ (pure rings), extended it to cycle-power skeletons $\vec{C}^{\,k}_m$, and obtained $E^{\mathrm{sc}}(n) \leq 5$ at $n = 47, 53, 59$ by transplanting a locally rigid cluster found by search. All three are subsumed numerically by Theorem 2.6: at $n = 25, 35, 49$ the cycle-power bound gave $5, 5, 7$ where the theorem gives $4, 4, 4$; at the three primes the transplant gave 5 where the theorem gives 4. The pure ring is the special case $c_a = 1$ of the family above, and G_n is exactly $\vec{C}_3$ blown up with truncated intra-class tournaments.

The transplant is worth keeping for its mechanism rather than its numbers. It is built around a **keystone cluster**: a satisfied vertex u of margin $+1$ together with *wing* vertices whose out-sets are exact copies of $N^+_1(u)$ plus the single arc back to u — out-set inclusion again, asymmetric and ring-free. We still have no general lemma saying when such a cluster embeds in a host of prescribed order with controlled excess (Problem 6).

3. An exact value, and the state of the lower bound

3.1. **A degenerate family where E is determined.**

Proposition 3.1. *Let $k \geq 1$ and $n = 2k + 1$. Every oriented graph G on n vertices with $\delta^+(G) \geq k$ satisfies $\mathrm{exc}(G) = n$. Hence $E_{\geq k}(2k+1) = E^{\mathrm{sc}}_{\geq k}(2k + 1) = 2k + 1$; in particular $E_{\geq 8}(17) = 17$.*

Proof. The arc count of an oriented graph on n vertices is at most $\binom{n}{2}$, and $\delta^+ \geq k$ forces it to be at least $nk = \binom{2k+1}{2}$. So equality holds throughout:

every pair is adjacent and every out-degree is exactly k, i.e. G is a regular tournament.

Fix v and let $w \in N^-(v)$. Suppose $w \notin N_2^+(v)$. For any $y \in N_1^+(v)$ we have $v \to y$; if $y \to w$ then $v \to y \to w$ is a path of length two and, since $w \notin N_1^+(v) \cup \{v\}$, this would put $w \in N_2^+(v)$. As G is a tournament, $w \to y$ instead. Hence $N_1^+(v) \subseteq N_1^+(w)$; and $w \to v$ gives $v \in N_1^+(w)$ while $v \notin N_1^+(v)$, so $N_1^+(v) \cup \{v\} \subseteq N_1^+(w)$ and $|N_1^+(w)| \geq k+1 > k$, contradicting regularity. Therefore $N^-(v) \subseteq N_2^+(v)$ and $d_2(v) \geq k$. Conversely $N_2^+(v) \subseteq V \setminus (N_1^+(v) \cup \{v\})$ gives $d_2(v) \leq n-1-k = k$. So $m(v) = 0$ at every vertex and $\mathrm{exc}(G) = n$. Finally $N_2^+(v) = N^-(v)$ puts every vertex within distance two of every other, so all these graphs are strongly connected and the two minima agree. The class is non-empty for every k — the circulant tournament on $\mathbb{Z}_{2k+1}$ with connection set $\{1, \dots, k\}$ is regular of out-degree k — so the minima are attained and the equalities hold rather than being vacuous. □

Proposition 3.1 is not new: it is the classical fact that a vertex of maximum score in a tournament reaches every other vertex within two steps, applied to a regular tournament where every vertex qualifies; the special case we need is [GKZ26, Lem. 2.2]. We state it because of what it replaces. Version 1.1 proved $E_{\geq 8}(17) \geq 3$ from a CP-SAT infeasibility run of 357 s, and a later run of 3 083 s at a higher cap pushed the same statement to ≥ 5. Both are true, and both are strictly weaker than the six lines above. We record this rather than quietly dropping it: the computations were sound, and the conclusion they reached was simply not the strongest available.

Remark 3.2 (why $n = 2k+1$ says nothing about anything else). The hypothesis is doing all the work, and dropping it by one collapses the value rather than lowering it slightly. At $n = 5$, $k = 2$ the oriented graph with out-sets $\{3,4\}, \{3,4\}, \{3\}, \{4\}, \{2\}$ is digon-free with $\delta^+ = 1$ and excess 3, not 5. So $E_{\geq 8}(17) = 17$ is an isolated rigid point: the degree bound pins the class down to a single excess value, and at $n = 18$ non-tournaments of low excess are immediately admissible. The large value is *not* evidence for a lower bound anywhere else.

3.2. **What is known on the lower side.** Nothing beyond the trivial $E_{\geq 8}(n) \geq 0$: no positive lower bound is known at any $n \geq 18$. The band $18 \leq n \leq 23$, which the family G_n does not reach, is not however without upper bounds. The blow-up family $\mathcal{F}$ of §4 has no member at $n = 18$ or $n = 19$, but its minima at the next four orders are

$$E_{\geq 8}(20) \leq 5, \quad E_{\geq 8}(21) \leq 6, \quad E_{\geq 8}(22) \leq 6, \quad E_{\geq 8}(23) \leq 6,$$

witnessed by blow-ups of the 5-vertex circulant tournament $C_5(1,2)$ with class sizes $(4,4,4,4,4)$, $(3,5,4,4,5)$, $(4,4,5,4,5)$ and $(4,5,4,5,5)$; the four graphs are published and re-scored by `make verify-band`, and `--full` re-derives that these are the minima over $\mathcal{F}$ and that $n = 18, 19$ admit no member. What is missing in the band is a lower bound and a working search instrument — the density forced by $\delta^+ \geq 8$ leaves almost no legal local moves. For $n \geq 24$, $3 + [3 \nmid n]$ is an upper bound only, and whether it is exact is open. Any bound of the form $E_{\geq 8}(n) \geq 2$ for all n is already a strengthening of SSNC restricted to this class; the case $c = 2$ is the restriction to $\delta^+ \geq 8$ of a conjecture of Dara, Francis, Jacob and Narayanan [DFJN22] that every sink-free oriented graph contains at least two Seymour vertices.

3.3. **Where the direct solver model reaches.** The CP-SAT model $M_{n,c}$ of Appendix A decides "exc $\leq c$" at $n = 17$ for $c = 2, 3, 4$ (357.0 s, 1 356.4 s, 3 083.0 s, all INFEASIBLE) and returns UNKNOWN at $n = 19$ and $n = 20$ for $c = 3$ within 1 800 s each, and at $18 \leq n \leq 22$ for $c = 2$ within 3 600 s each. The tournament-restricted variant of "exc ≤ 2" returned twelve consecutive UNKNOWNs at one hour each over $17 \leq n \leq 28$. After Proposition 3.1 the $n = 17$ rows are of interest only as a measurement of the model: they consumed up to an hour to establish something a paragraph settles, which is the honest calibration of how far this approach reaches. It does not reach the orders where the question is live.

3.4. **Provenance of the solver logs.** The logs are not uniform, and a reader comparing claims should know which tier each rests on. Counted field by field over every row:

log	rows	model hash	solver version	workers
`gkz82_results.jsonl`	25	0	0	0
`excess2_results.jsonl`	8	1	1	0
`excess3_probe_results.jsonl`	4	4	4	4

The GKZ scan records conjecture, k, n, two model flags, status and wall time — seven fields, except for the rows at $n = 7, 8, 9$, which predate one of the flags and carry six. The exc ≤ 2 log carries four fields on seven of its eight rows and seven on the remaining one, which alone has a model hash and solver version. The reachability probe carries thirteen fields on every row, including model hash, solver version and worker count; none of the three logs records a solver seed. All models are in the repository and can be re-run; only the third log pins down the binary that produced it.

Two instances of a status changing on re-run are kept in the logs, and they are not the same phenomenon. In the GKZ scan, $n = 29$ was UNKNOWN at 14 400 s and INFEASIBLE at 29 506 s: a budget was exhausted and roughly twice the budget decided it. In the exc ≤ 2 log, $n = 17$ was UNKNOWN at 900 s and then INFEASIBLE at 218.9 s — the deciding run was four times *faster* than the run that gave up, so the budget was not what was missing, and the rows do not record what differed. The second case is the sharper illustration of a point this note makes elsewhere: a budget-exhausted UNKNOWN is evidence in neither direction, and does not even reliably indicate that an instance is hard.

3.5. **A finite verification of a conjecture of Guo, Kang and Zwaneveld.** [GKZ26, Conj. 8.2] predicts a structural property of Seymour-tight orientations. We record a finite verification, which belongs here because the note, not the repository README, is the statement of what this work asserts; version 1.1 carried this result only in the README, which is the same defect this version repairs elsewhere.

Proposition 3.3 (computer-assisted)**.** *For $k = 3$, Conjecture 8.2 of* [GKZ26] *has no counterexample on n vertices for any $7 \leq n \leq 30$.*

The verification is a CP-SAT infeasibility scan, one instance per order; the hardest, $n = 29$, was first abandoned at a four-hour budget and decided at 29 506 s on a longer re-run, and both rows are kept in the log. The scan is non-vacuous: hypothesis graphs, i.e. objects satisfying the conjecture's premise, were exhibited at every scanned order, so the infeasibility is not that of an empty hypothesis. The conclusion is conditional on the encoding and on solver soundness in the sense of Appendix A, and this scan rests on the *thinnest* log in the repository — see §3.4: its rows carry neither a model hash nor a solver version. Of the results in this note it is the one whose record we would most want strengthened before it is relied upon.

4. A class-limited partial lower bound

The only lower-bound-shaped statement we can prove beyond Proposition 3.1 holds inside a family we define, and we are explicit that it says nothing outside it.

Theorem 4.1 (computer-assisted; restricted to $\mathcal{F}$, see Appendix B)**.** *Let $\mathcal{F}$ be the family of capped-tournament blow-ups $H[n; c]$ over a strongly connected oriented quotient H on m vertices, $3 \leq m \leq 6$, whose minimum out-degree is at least* 8 *(Appendix B gives the construction in full). Then every $G \in \mathcal{F}$ has $\mathrm{exc}(G) \geq 3$, and $\mathrm{exc}(G) = 3$ implies $3 \mid n$. The quantification is over the members of $\mathcal{F}$, with no bound on n.*

This is not a lower bound on $E_{\geq 8}(n)$. Most oriented graphs are not blow-ups of this kind, and the statement says nothing whatsoever about anything outside $\mathcal{F}$; read outside $\mathcal{F}$ it would be a strengthening of SSNC itself. The unswept range is quotients on $m \geq 7$ vertices and every construction that is not a capped-tournament blow-up — which is almost all of them.

Two points about the statement were mis-stated in earlier drafts and are corrected here. First, $3 \leq m \leq 6$ and not $m \leq 6$: there is no strongly connected oriented graph on two vertices, and on one vertex the sole graph has no arcs and fails the degree condition, so $m = 1, 2$ contribute nothing; the quotients enumerated number $4\,313 = 1 + 4 + 76 + 4\,232$ for $m = 3, 4, 5, 6$. Second, $\mathcal{F}$ is *not* empty below $n = 24$: its smallest member has $n = 20$, obtained from the 5-vertex circulant tournament with all five classes of size 4, where every class out-sum is $4 + 4 = 8$. That 20 is optimal follows by minimising $\sum_a n_a$ over all orientations of the quotient, which returns $24, 24, 20, 20$ for $m = 3, 4, 5, 6$. The earlier claim of emptiness came from extrapolating the $m = 3$ case, where each quotient vertex has out-degree one and the degree condition forces *every* class to have at least 8 vertices.

5. Measurements

5.1. **Table.** The table is machine-generated from the repository data (`note/make_table.py`); hashes are canonical-adjacency SHA-1 prefixes from the witness manifest.

Table 1. Surveyed orders: the theorem against the constructions it supersedes.

n	$3 + [3 \nmid n]$	earlier construction	fresh search	best stored witness
24	3	3 (ring m=3,k=1,t=8)	-	`2517130b`
25	4	5 (ring m=5,k=2,t=5)	12	`150003fd`
27	3	3 (ring m=3,k=1,t=9)	-	`5b5ae497`
30	3	3 (ring m=3,k=1,t=10)	3	`feba34e7`
35	4	5 (ring m=5,k=2,t=7)	7	`ca62560c`
40	4	4 (ring m=4,k=1,t=10)	23	`9cb88021`
45	3	3 (ring m=3,k=1,t=15)	21	`abd86a45`
47	4	5 (keystone transplant)	9	`ccc19f2e`
48	3	3 (ring m=3,k=1,t=16)	13 (seed-free)	`d43ca1db`
49	4	7 (ring m=7,k=2,t=7)	6	`db2e4026`
50	4	5 (ring m=5,k=1,t=10)	5 (20 seed-free)	`bce286aa`
51	3	3 (ring m=3,k=1,t=17)	-	`e5d2d5eb`
53	4	5 (keystone transplant)	5	`8562bd98`
57	3	3 (ring m=3,k=1,t=19)	-	`d02d73a9`
59	4	5 (keystone transplant)	7	`d90d86c3`
60	3	3 (ring m=3,k=1,t=20)	33	`44b4bca8`
75	3	3 (ring m=3,k=1,t=25)	3	`63346a8f`

Table 2. The measured witnesses of `data/sonar_best/`, summarised.

quantity	value
orders covered	24..100, no gaps
stored witnesses	77 (one per order)
excess $= 3 + [3 \nmid n]$	77 of 77 orders
orders disagreeing	none
min out-degree observed	8 (n=24) .. 33 (n=99)
strongly connected	77 of 77 orders

5.2. **Search attainment is not the floor.** This is the methodological result of the project, and the reason for the terminology fixed in §1. Search attainment overstated the best available upper bound in every discrepancy we recorded, and in each one an explicit construction — never a further search — settled the comparison. The two failures below differ in kind, not in direction.

Failure by non-attainment. A pre-registered control — fresh search on the composites $n = 48, 50$ with all construction seeds withheld — returned 13 and 20 against construction values 3 and 5 at the same orders. The seed, not the search, had been carrying the composite values. Above the surveyed band the gap persists but is narrower: over the 50 orders $101 \le n \le 150$ the best stored witness matches $3 + [3 \nmid n]$ at 40 of them and stands above it at 10, attaining 5 or 6 against a construction value of 4, the largest shortfall being $+2$. Those witnesses are published with this note (`data/sonar_high/`) and the sentence is re-derived from them by `make verify-m1-high`; none of them falls *below* the construction value.

Failure by false pattern inference. At the primes $n = 59$ and $n = 61$ an earlier search settled on excess 7, and this was read as a "floor = ring length +1" pattern at primes. It was a basin artefact; the pre-registered hypothesis is now rejected, the construction gives 4 at both orders, and witnesses of excess 4 at both are stored. Note the shape of this error: the search did not merely fail to reach the bound, it produced a value consistent enough across two orders to look like a law.

δ-invariance, restated. Version 1.1 reported that the best value observed at $n = 50$ was unchanged when the degree bound was raised to $\delta^+ \ge 9, 10$. The observation survives in a stronger and less interesting form: by Remark 2.7 the construction gives 4 at $n = 50$ for every $\delta \le 16$, so invariance across $\delta \in \{8, 9, 10\}$ is a property of the family rather than a discovery about the landscape.

6. Structure of two witnesses at excess 5

Two search-discovered witnesses, analysed in version 1.1 as the best known at their orders. They are *not* the witnesses listed at $n = 50$ and $n = 53$ in Table 1: those are the stored witnesses attaining the theorem value 4, whereas the two graphs below have excess 5 and are kept because this section is about them.

- $n = 50$ (`17cd7dc8`): five tight survivors forming a **pentagon ring** in which each survivor's first neighbourhood contains the next survivor and its second neighbourhood the one after; the sensitivity matrix over all valid single arc-flips is diagonal.
- $n = 53$ (`dc922e89`): a **keystone cluster** with margins $(+1, 0, 0, 0)$ and the wings' out-sets equal to the keystone's; the only non-diagonal row of the sensitivity matrix is at the keystone, where killing it raises both wings.

Different architecture and different margin economy, yet identical local dynamics: neither admits an improving single flip, both lie on large neutral plateaus, and the keystone's apparent $+1$ slack survives an exhaustive two-flip attack over all 171 288 valid coordinated pairs *restricted to the cluster neighbourhood* — not an unrestricted depth-2 search.

Remark 6.1 (a reading withdrawn). Version 1.1 offered this as "consistent with, though not implying, a global counting obstruction rather than a local structural one". That reading is withdrawn. Theorem 2.6 gives 4 at $n = 50$ and 4 at $n = 53$, so both witnesses sit above the constructive value and their rigidity is the rigidity of a search basin, not of an optimum. The measurements remain valid as measurements of those two graphs.

Remark 6.2 (Pisa structure)*.* All witnesses above are Pisa graphs whose underlying undirected graphs are irregular and far from K_n minus a matching; together with small blow-ups of directed cycles they show that the classification of underlying graphs of Pisa graphs is substantially richer than conjectured in [Hal26, Conj. 5.1]. Explicit counterexamples were communicated to the author (private communication, July 2026).

7. Open problems

(1) **Is** $3 + [3 \nmid n]$ **exact?** Equivalently, is $E_{\geq 8}(n) \geq 3$ for all n, with equality when $3 \mid n$? We have no lower-bound evidence at any $n \geq 18$ — only Theorem 4.1, which is class-limited, and the fact that a broad sweep never went below the construction.

(2) **The band** $18 \leq n \leq 23$**.** No construction, no lower bound, no working search instrument. The smallest genuinely unexplored gap.

(3) **Beyond the blow-up class.** Theorem 4.1 closes $\mathrm{exc} \leq 2$ inside $\mathcal{F}$. What happens for quotients on $m \geq 7$ vertices, and for constructions that are not blow-ups at all?

(4) **A certificate for infeasibility.** Every INFEASIBLE result here rests on an encoding argument and on solver soundness. A proof-logging solver would remove the second dependency. Proposition 3.1 removes it at $n = 17$ by dispensing with the solver.

(5) **Global counting.** Find a double-counting or matrix-analytic proof of any bound $E_{\geq 8}(n) \geq 2$. The matrix S_D of [GKZ26] expresses margins as row sums and excess as the positive part of $S_D\mathbf{1}$.

(6) **Transplant lemma.** Formulate and prove a general lemma specifying when the keystone cluster of §2.3 embeds in a host of given order with excess bounded by a constant.

(7) **Tournaments with exactly two tight vertices.** Does a tournament with $\delta^+ \geq 8$ and exactly two satisfied vertices, both tight, exist? Twelve one-hour CP-SAT attempts at $17 \leq n \leq 28$ returned UNKNOWN, and the $n = 17$ case is settled negatively by Proposition 3.1 before any computation.

8. Methods and reproducibility

Search: island-model hill climbing with plateau acceptance, on CPU. Version 1.1 reported LLM-driven program evolution; that pipeline was retired when the classical search reached the same values faster and over a wider range of orders, and the witnesses shipped with this version come from the classical search. Verification: independent implementations as described in §1; every claim maps to a `make verify-*` target. Pre-registration: search predictions were committed to the repository before the corresponding runs.

Division of labour: direction and decisions by the human author; supervision, literature and independent verification, and implementation, search and solver engineering, by Claude (Anthropic). All witness-based numerical claims are independently machine-verifiable from the accompanying repository; the computer-assisted results carry the limitations stated in Appendix A and §3.4.

Repository: `https://github.com/42tahara/seymour-excess`. DOI: `https://doi.org/10.5281/zenodo.21497370` (software); `https://doi.org/10.5281/zenodo.21497888` (this note). Both are concept DOIs and resolve to the newest version.

Appendix A. The CP-SAT model $M_{n,c}$

We describe the model behind §3.3 completely, so that it can be checked against the paper alone; the implementation is `experiments/delta8/excess2_search.py` and the reachability ladder of §3.3 calls the same builder with a larger c.

Variables. For ordered pairs $i \neq j$ of $[n]$, Boolean *arc* variables a_{ij}; for ordered pairs $v \neq w$, Boolean *distance-two* variables s_{vw}; for each v, an integer $e_v \in \{0, \dots, n\}$.

Constraints.

(O) $a_{ij} + a_{ji} \leq 1$ for all $i < j$ (orientation);
(D) $\sum_{j \neq i} a_{ij} \geq 8$ for all i (out-degree);
(S1) $s_{vw} + a_{vw} \leq 1$ (N_1^+ and N_2^+ disjoint);
(S2) $a_{vy} + a_{yw} - a_{vw} - s_{vw} \leq 1$ for all distinct v, y, w;
(E) $e_v \geq \sum_{w \neq v} s_{vw} - \sum_{j \neq v} a_{vj} + 1$;
(C) $\sum_v e_v \leq c$;
(B) symmetry breaking: $\sum_j a_{0j} \geq \sum_j a_{ij}$ for all i, and $a_{0,j} \geq a_{0,j+1}$ for $1 \leq j \leq n-2$.

Lemma A.1 (graph-to-model direction)**.** *If an oriented graph G on n vertices with $\delta^+(G) \geq 8$ and $\operatorname{exc}(G) \leq c$ exists, then $M_{n,c}$ is feasible.*

Proof. Relabel G so that vertex 0 has maximum out-degree and its out-neighbours are $1, \dots, d_1(0)$; this meets (B). Set $a_{ij} = 1$ iff $i \to j$; (O) holds since G is oriented and (D) since $\delta^+ \geq 8$. Set $s_{vw} = 1$ iff $\operatorname{dist}^+(v, w) = 2$; (S1) holds by disjointness and (S2) because a two-path with $a_{vw} = 0$ witnesses distance exactly two. Then $\sum_w s_{vw} = d_2(v)$, so $e_v = \max(0, d_2(v) - d_1(v) + 1)$ satisfies (E), and (C) is $\operatorname{exc}(G) \leq c$. □

Hence INFEASIBLE implies no such graph exists.

Strong connectivity is not encoded. The model contains **no** strong-connectivity constraint, and Lemma A.1 assumes none: *every* oriented graph with $\delta^+ \geq 8$ and $\operatorname{exc} \leq c$ yields a feasible assignment. Consequently INFEASIBLE rules out the larger class, and the bound for the strongly connected subclass follows a fortiori. Omitting a constraint is a relaxation, and a relaxation can only turn infeasible instances feasible, never the reverse.

Feasibility direction. The second deliberate relaxation: s_{vw} may be 1 without a witnessing two-path (it is forced upward by (S2), never downward), so a *feasible* assignment need not describe a real graph of low excess. Every FEASIBLE output of this model is re-verified exactly on the extracted adjacency matrix before being believed; INFEASIBLE outputs need no such step.

Limitations. The independent-implementation protocol verifies *witnesses*; it does not independently verify *infeasibility*, which rests on the encoding above and on solver soundness. See §3.4 for what each log records.

Appendix B. The capped-tournament blow-up family

Definition. Let H be a strongly connected oriented graph on m vertices, $3 \le m \le 6$. For a vector of positive integers $(n_1, \dots, n_m)$ and a vector of integers $(c_1, \dots, c_m)$ with $0 \le c_a \le n_a$, define the **capped-tournament blow-up** $G = H[n; c]$ as follows. The vertex set is the disjoint union of classes $C_1, \dots, C_m$ with $|C_a| = n_a$, whose elements are written $u_{a,0}, \dots, u_{a,n_a-1}$. The arcs are of these two kinds only: **(R)** whenever H has $a \to b$, then $u_{a,j} \to$ every vertex of C_b, for every j; **(T)** $u_{a,j} \to u_{a,l}$ for every pair (j, l) with $l < \min(j, c_a)$. Put $n = \sum_a n_a$. Write $\mathcal{F}$ for the set of all such G whose minimum out-degree is at least 8.

Claim — about $\mathcal{F}$ only; we do not quantify over arbitrary oriented graphs. Every $G \in \mathcal{F}$ satisfies $\operatorname{exc}(G) \ge 3$, and if $\operatorname{exc}(G) = 3$ then $3 \mid n$. The quantification is over the members of $\mathcal{F}$, with no bound on n.

This is not a lower bound on $E_{\ge 8}(n)$. Since most oriented graphs are not blow-ups of the above kind, this claim says nothing whatsoever about anything outside $\mathcal{F}$.

The proof is computer-assisted. What is proved by hand is everything up to a finite decision problem; what a solver contributes is the decision itself, on explicitly bounded instances. This appendix gives the reduction in full, so that the only thing taken on trust is the solver, in the same sense as Appendix A.

B.1 Per-class arithmetic. Fix H and write $A(a)$ for the out-neighbourhood of a in H, $B(a) = \bigcup_{b \in A(a)} A(b)$ for its two-step reach, and $C(a) = B(a) \setminus A(a)$. Put

$$S_1(a) = \sum_{b \in A(a)} n_b, \qquad S_2(a) = \sum_{c \in C(a)} n_c.$$

Lemma B.1. *$G = H[n;c]$ is an oriented graph iff H is; G is strongly connected iff H is (for $m \geq 2$); and* $\deg^+(u_{a,j}) = S_1(a) + \min(j, c_a)$*, so* $\delta^+(G) = \min_a S_1(a)$*, independently of the caps.*

Proof. No loops, since (T) needs $l < \min(j, c_a) \leq j$. A digon would need either $a \to b$ and $b \to a$ in H, or two (T) arcs in opposite directions inside one class, and (T) arcs decrease the index; (R) and (T) arcs never join the same pair. The out-degree formula is immediate from the definition, and is minimised at $j = 0$. For connectivity, the class partition is a congruence: reachability in G projects onto reachability in H, and if H is strongly connected then from any $u \in C_a$ one reaches all of C_b for some $b \in A(a)$, hence, following a walk in H, every class entirely, and re-enters C_a along an (R) arc. $\square$

Lemma B.2. $N_2^+(u_{a,j}) = \bigcup_{c \in C(a)} C_c$*, so* $d_2(u_{a,j}) = S_2(a)$*, independently of j and of every cap. Consequently*

$$m(u_{a,j}) \;=\; \Delta_a - \min(j, c_a), \qquad \Delta_a \;:=\; S_2(a) - S_1(a).$$

Proof. Write $p = \min(j, c_a)$, so $N_1^+(u_{a,j}) = \bigcup_{b \in A(a)} C_b \cup P$ with $P = \{u_{a,0}, \ldots, u_{a,p-1}\}$. An intra-class target $u_{a,l}$ with $l < p \leq c_a$ has $N_1^+(u_{a,l}) = \bigcup_{b \in A(a)} C_b \cup \{u_{a,0}, \ldots, u_{a,l-1}\}$, which is contained in $N_1^+(u_{a,j})$ since $l-1 < p$; the (T) arcs therefore contribute nothing new. A vertex $u_{b,l}$ with $b \in A(a)$ contributes $\bigcup_{c \in A(b)} C_c$ together with a subset of C_b. Unioning and deleting $N_1^+(u_{a,j}) \cup \{u_{a,j}\}$, which contains $\bigcup_{b \in A(a)} C_b$, leaves $\bigcup_{c \in B(a) \setminus A(a)} C_c$. Finally $a \notin B(a)$: otherwise $a \to b \to a$ would be a digon of H. So the deleted set, which lives in C_a, is disjoint from what remains. The margin formula follows from $d_1(u_{a,j}) = S_1(a) + \min(j, c_a)$. $\square$

Lemma B.2 is the free out-degree mechanism of Lemma 2.1 again: the intra-class targets are chosen so that their out-sets already lie inside the observer's first neighbourhood.

B.2 The class-excess function, and why it depends on two variables. By Lemma B.2 the contribution of class a to $\mathrm{exc}(G)$ is

$$X(a) \;=\; \sum_{j=0}^{n_a - 1} \max\bigl(0,\, \Delta_a - \min(j, c_a) + 1\bigr),$$

which depends on the class only through (Δ_a, n_a, c_a) — the positions $j \geq c_a$ all contribute $\max(0, \Delta_a - c_a + 1)$ and the positions $j < c_a$ contribute $\max(0, \Delta_a - j + 1)$.

Lemma B.3. $\min_{0 \leq c_a \leq n_a} X(a) = g(\Delta_a, n_a)$*, where $g(\Delta, \nu) = 0$ for $\Delta \leq -1$ and, for $\Delta \geq 0$,*

$$g(\Delta, \nu) \;=\; \sum_{i=0}^{t-1} (\Delta + 1 - i) \;=\; t(\Delta + 1) - \tbinom{t}{2}, \qquad t = \min(\nu,\, \Delta + 1).$$

The minimum is attained at $c_a = \min(n_a, \max(0, \Delta_a + 1))$ *and also at* $c_a = n_a$*. Hence a lower bound on* exc *proved at the minimising cap holds for every cap, and the cap vector need not be enumerated.*

Proof. Increasing c_a by one replaces each contribution $\max(0, \Delta_a - c_a + 1)$ at a position $j \geq c_a$ by $\max(0, \Delta_a - c_a)$, which is not larger; so $X(a)$ is non-increasing in c_a and saturates once $c_a \geq \Delta_a + 1$, where every further position contributes 0. Evaluating at saturation gives the displayed sum with $t = \min(n_a, \Delta_a + 1)$ terms. □

Two monotonicity facts make the search space finite.

Lemma B.4. *(i) For* $\Delta \geq 0$, $g(\Delta, \nu) \geq \Delta + 1$*; hence if the total excess is at most* E *then* $\Delta_a \leq E - 1$ *for every* a*. (ii)* $g(\Delta, \cdot)$ *is constant for* $\nu \geq \Delta + 1$*. Consequently, for a fixed budget* E*, the pairs* (Δ_a, n_a) *fall into finitely many regimes, namely* $\Delta \leq -1$ *with* $\nu \geq 1$ *(cost* 0*) and, for each* $0 \leq \Delta \leq E - 1$*, the cases* $\nu = 1, \ldots, \nu = \Delta$ *and* $\nu \geq \Delta + 1$*; only those of cost at most* E *need be kept.*

Proof. (i) The summands are $\Delta + 1, \Delta, \ldots, \Delta + 2 - t$ with $t \geq 1$, all positive, so $g \geq \Delta + 1$ when $t = 1$ and $g \geq (\Delta + 1) + \Delta \geq \Delta + 1$ when $t \geq 2$. (ii) t saturates at $\Delta + 1$. □

For $E = 3$ the surviving list is exactly

NEG	$\Delta \leq -1$	$\nu \geq 1$	cost 0
D0M	$\Delta = 0$	$\nu \geq 1$	1
D1S1	$\Delta = 1$	$\nu = 1$	2
D1M	$\Delta = 1$	$\nu \geq 2$	3
D2S1	$\Delta = 2$	$\nu = 1$	3

Note that the unit of enumeration is the pair $(\Delta_a, \min(n_a, \Delta_a + 1))$ and not Δ_a alone: the degree condition binds $S_1(a) = \sum_{b \in A(a)} n_b$, not n_a, so single-vertex classes are admissible and do occur. Enumerating Δ alone would miss them, and 166 of the feasible instances at budget 4 do place a class in the regime D1S1, which exists only at $n_a = 1$.

B.3 The linear model. Let $M_1, M_2 \in \{0, 1\}^{m \times m}$ be the incidence matrices of A and of C, so $(M_1 s)_a = S_1(a)$ and $(M_2 s)_a = S_2(a)$ for a size vector $s = (n_1, \ldots, n_m)^\top$, and put $N = M_2 - M_1$. Then $\Delta = Ns$ is a *single linear map* of s — not a piecewise one — which is what allows the question to be inverted. Fixing an assignment of regimes to classes turns the constraints into a rational polyhedron

$$P \;=\; \{\, s \in \mathbb{R}^m \;:\; s \geq \mathrm{lo},\; s \leq \mathrm{hi},\; M_1 s \geq 8,\; (Ns)_a \text{ in its regime's range} \,\},$$

and $n = \mathbf{1}^\top s$. Every constraint has the form $(\text{row}) \cdot s \leq \text{const}$, and P lies in the non-negative orthant, so P is pointed.

B.4 Removing the bound on n. A solver needs bounded domains. Rather than *assume* a bound on n, one is *derived* per instance.

Lemma B.5 (bound derivation)**.** *Let $P = \mathrm{conv}(V)+\mathrm{cone}(R)$ be the Minkowski–Weyl decomposition of P, with V its vertices and R its extreme rays taken integral and primitive, and set*

$$B \;=\; \max_{q\in V} \mathbf{1}^\top q \;+\; 3\sum_{r\in R} \mathbf{1}^\top r.$$

If P contains an integer point s with $\mathbf{1}^\top s \equiv \rho \pmod 3$, then it contains one with $\mathbf{1}^\top s \le B$ and the same residue ρ.

Proof. Write $s = q + \sum_i \mu_i r_i$ with $q \in \mathrm{conv}(V)$ and $\mu_i \ge 0$, and put $s' = s - \sum_i 3\lfloor \mu_i/3 \rfloor r_i$. The subtracted coefficients are non-negative and at most μ_i, so $s' \in P$; the r_i are integral, so s' is integral; each subtracted term has coordinate sum $3\lfloor \mu_i/3 \rfloor\, \mathbf{1}^\top r_i$, a multiple of 3, so $\mathbf{1}^\top s' \equiv \mathbf{1}^\top s \pmod 3$; and the surviving coefficients lie in $[0, 3)$, so $\mathbf{1}^\top s' \le \max_{q\in V} \mathbf{1}^\top q + 3\sum_i \mathbf{1}^\top r_i = B$. □

V and R are computed exactly, in integer arithmetic, by the double-description method on the homogenisation of P. If V is empty then P is empty over the rationals and a fortiori over the integers.

Lemma B.6 (soundness)**.** *Fix a quotient H and a regime assignment of total cost at most E. Deciding, for each residue $\rho \in \{0, 1, 2\}$, whether P contains an integer point with $\mathbf{1}^\top s \le B$ and $\mathbf{1}^\top s \equiv \rho$ decides, for that instance, whether a member of $\mathcal{F}$ realising that regime assignment, with excess at most E and $n \equiv \rho \pmod 3$, exists at any n whatsoever. Ranging over all quotients and all regime assignments of cost at most E therefore decides the question for $\mathcal{F}$.*

Proof. Soundness in one direction is Lemma B.3: a size vector in P yields a member with excess $\sum_a g(\Delta_a, n_a) \le E$ at the minimising cap. In the other direction, a member with excess $\le E$ has, by Lemma B.4, every class in one of the finitely many regimes, so its size vector lies in P for the corresponding assignment; and by Lemma B.5 if any such integer point exists at residue ρ then one exists below B. So the bounded search is exhaustive over all n. □

B.5 The conserving quotients, by hand. One family needs no computation. Column c of N sums to $|\{a : c \in C(a)\}| - \deg^-_H(c)$, so $\mathbf{1}^\top N = 0$ — call H *conserving* — iff $\sum_a \Delta_a = 0$ for every size vector.

Proposition B.7. *If H is conserving then every $G = H[n; c]$ has $\mathrm{exc}(G) \ge m$, for every size vector and every cap vector.*

Proof. Let $S = \{a : \Delta_a \ge 0\}$. Since $\sum_a \Delta_a = 0$ and each $a \notin S$ has $\Delta_a \le -1$, we get $\sum_{a\in S} \Delta_a = -\sum_{a\notin S} \Delta_a \ge m - |S|$. Each $a \in S$ contributes at least $\Delta_a + 1$ by Lemma B.4(i), so $\mathrm{exc} \ge |S| + \sum_{a\in S} \Delta_a \ge |S| + (m - |S|) = m$. □

Of the 4 313 strongly connected oriented quotients with $3 \le m \le 6$ exactly 7 conserve $(1, 1, 2, 3$ for $m = 3, 4, 5, 6)$; for $m \ge 4$ this already excludes excess ≤ 3 outright, and for $m = 3$ it is the classical hand computation for the directed triangle.

B.6 The computation, and what it returned. Enumerating the 4 313 quotients and, for each, every assignment of the regimes of B.2 with total cost at most 3, gives 385 695 instances. Of these 385 622 are empty already over the rationals; none is undecided; 73 are feasible. The largest derived bound B over all instances is 222. Every feasible instance has excess exactly 3 and $n \equiv 0 \pmod 3$, with profile three D0M classes and the rest NEG; no instance with excess 0, 1 or 2 is feasible for any n. Raising the budget to 4 gives 1 052 936 instances, 1 051 964 of them empty over the rationals, none undecided, and 972 feasible (73 with excess 3, 899 with excess 4), with largest derived bound $B = 224$; excess 4 occurs in all three residues, and the smallest n carrying excess ≤ 4 anywhere in the family is 24. Both runs are published as `data/blowup/blowup_inverse_results.jsonl` and `..._e4.jsonl`: every figure quoted in this subsection is a field of the summary row of one of those two files, except for the four cross-checks of the next paragraph, whose figures come from the output of `make verify-t9`; the per-instance rows are included so that the residue pattern can be read off directly.

Four bounded cross-checks accompany the computation (`make verify-t9`), each of which would fail loudly if the reduction above were wrong: the 73 feasible size vectors are expanded into explicit graphs and re-scored (73/73 agree); 2 321 385 blow-ups at small orders are evaluated exhaustively, finding no excess ≤ 3 with $3 \nmid n$; all 357 irreducible elements of the recession cones found within the search box $[0, 10]$ are checked to have coordinate sum $\equiv 0 \pmod 3$; and all 146 base models at residues 1 and 2 are confirmed infeasible.

B.7 What is not covered. Quotients on $m \ge 7$ vertices, and every construction that is not a capped-tournament blow-up in the sense of the definition above — in particular every blow-up whose intra-class structure is not a truncated transitive tournament. That is almost all oriented graphs, which is why the result is stated as a property of $\mathcal{F}$ and not as a lower bound.

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